

RIVERSIDE CANCER CENTER Leukemia Service

February 8, 2024

Dr. Jennifer Walsh Internal Medicine Associates 4500 Medical Plaza Drive Riverside, CA 92501

RE: Michael Brown DOB: 06/18/1955 MRN: AML005

Dear Dr. Walsh,

I am writing to provide you with a comprehensive update on our mutual patient, Mr. Michael Brown, who continues to do remarkably after his initial diagnosis of acute myeloid leukemia.

INITIAL PRESENTATION AND DIAGNOSIS:

Mr. Brown initially presented to your office in August 2021 with a 6-week history of progressive fatigue, night sweats, and a 15-pound unintentional weight loss. You astutely noted his concerning CBC (WBC 67,000 with 45% circulating blasts, hemoglobin 7.8 g/dL, platelets 42,000) and promptly referred him to our service.

His bone marrow biopsy on 08/14/2021 revealed 72% myeloblasts with intermediate-risk cytogenetics (normal karyotype). Comprehensive molecular profiling including a 54-gene NGS panel revealed no targetable mutations (specifically negative for FLT3, NPM1, CEBPA, IDH1/2, and TP53). He was classified as intermediate risk by ELN 2022 criteria.

TREATMENT COURSE:

Induction Therapy (August-September 2021): Mr. Brown was admitted on 08/21/2021 for standard 7+3 induction (cytarabine 100mg/m² continuous infusion days 1-7, daunorubicin 60mg/m² days 1-3). His BSA of 1.95m² required dose modification for cytarabine due to our pharmacy's maximum daily dose policy.

His induction course was complicated by:

- Febrile neutropenia (day 9-18) treated with cefepime, ultimately growing *Enterococcus faecium* requiring linezolid
- Grade 2 mucositis requiring PCA
- Tumor lysis syndrome managed with rasburicase
- Prolonged thrombocytopenia requiring 18 platelet transfusions

Day 14 marrow showed aplasia without residual leukemia. Count recovery began day 26 with ANC >500 achieved on day 29.

Day 30 Assessment: Restaging bone marrow (09/20/2021) showed complete remission with incomplete count recovery (CRi):

- 2% blasts (normal <5%)

- ANC 780
- Platelets 68,000
- MRD by flow cytometry: 0.08% abnormal myeloid cells

Consolidation Therapy (October 2021-February 2022):

Given his intermediate risk status, age 66, and excellent performance status, we proceeded with consolidation chemotherapy rather than immediate transplant.

Cycle 1 (October 2021): High-dose cytarabine (HiDAC) 3g/m² q12h days 1,3,5. Complicated by mild confusion (cytarabine neurotoxicity) that resolved with dose reduction to 1.5g/m² for remaining doses.

Cycle 2 (December 2021): HiDAC 1.5g/m² (dose-reduced) q12h days 1,3,5. Tolerated well with expected cytopenias.

Cycle 3 (January 2022): HiDAC 1.5g/m² q12h days 1,3,5. Required 7-day admission for neutropenic fever, cultures negative.

Cycle 4 (February 2022): Final HiDAC cycle completed outpatient with G-CSF support.

RELAPSE AND SALVAGE THERAPY (November 2022):

Unfortunately, routine surveillance labs in November 2022 showed rising WBC to 24,000 with 35% blasts. Bone marrow confirmed morphologic relapse with 52% blasts, unchanged molecular profile.

We discussed options including:

1. Clinical trial with novel FLT3 inhibitor (though he was FLT3 negative)
2. FLAG-IDA salvage chemotherapy
3. Venetoclax-based therapy
4. Supportive care only

Mr. Brown opted for FLAG-IDA (fludarabine, cytarabine, idarubicin, G-CSF) given his prior good response to intensive therapy. He achieved CR2 after one cycle (December 2022).

ALLOGENEIC STEM CELL TRANSPLANT (March 2023):

Given his CR2 status and available 10/10 matched unrelated donor (28-year-old male), Mr. Brown proceeded to reduced-intensity allogeneic SCT on 03/15/2023 following fludarabine/melphalan conditioning.

Post-transplant course:

- Day +30: Full donor chimerism (>95%)
- Day +60: Grade 1 skin GVHD managed with topical steroids
- Day +100: Tapered off tacrolimus
- Day +180: Remains in CR2, 100% donor chimerism

CURRENT STATUS (February 2024):

Mr. Brown is now day +330 post-transplant and continues in complete remission. His most recent evaluation (01/30/2024) showed:

Performance Status: ECOG 0 - fully active, back to playing golf twice weekly

Physical Examination: Essentially normal except for mild hyperpigmentation on forearms (resolved GVHD)

Laboratory Studies:

- CBC: WBC 5,800 (normal differential), Hgb 13.2 g/dL, Platelets 188,000
- Chemistry: Creatinine 1.0 mg/dL, LFTs normal
- Chimerism: 100% donor in all lineages
- Bone marrow (01/15/2024): Normocellular with trilineage hematopoiesis, no evidence of AML, MRD negative

Current Medications:

- Acyclovir 400mg BID (through month 12 post-SCT)
- Vitamin D 2000 units daily
- No immunosuppression since October 2023

LATE EFFECTS MONITORING:

We continue to monitor for late transplant effects:

- Pulmonary function tests q6 months (most recent normal)
- Annual dermatology evaluation
- Cardiovascular risk assessment (recent echo EF 60%)
- Immunizations being updated per post-transplant guidelines
- Sexual health: mild erectile dysfunction managed with sildenafil

PSYCHOSOCIAL:

Mr. Brown has done remarkably well psychologically. He attended our survivorship support group for 6 months and now serves as a peer mentor for newly diagnosed AML patients. His wife reports he has fully resumed his previous activities including woodworking and traveling (recent trip to Hawaii for 30th anniversary).

PROGNOSIS AND FOLLOW-UP PLAN:

With CR2 maintained for >10 months post-transplant and 100% donor chimerism, Mr. Brown's prognosis is cautiously optimistic. The risk of relapse decreases significantly after the first year post-transplant. Five-year overall survival for patients in his situation approaches 40-50%.

We will continue to see him:

- Monthly through month 12 post-SCT (March 2024)
- Every 3 months year 2
- Every 6 months years 3-5
- Annually thereafter

Please continue his routine health maintenance including:

- Colonoscopy (due this year)
- Annual PSA screening
- Cardiovascular risk management
- Pneumococcal and influenza vaccines (coordinate with us regarding timing)

Thank you for your excellent care of Mr. Brown throughout his journey. His positive outcome reflects the importance of early recognition and prompt referral that you provided. Please don't hesitate to contact me at 555-0147 with any questions or concerns.

With warm regards,

Katherine Roberts, MD Director, Leukemia Service Riverside Cancer Center

cc: Michael Brown Transplant Coordinator Survivorship Clinic